

Simplify Algebraic Expressions

Lesson 6-7

Name: _____

Date: _____

Class: _____

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.


$$3x + 2x = 5x$$

same variable → combine

5x and 3x are like terms (both x); 5x and 5y are NOT (different variables); 2x and 2x² are NOT (different powers)

Like Terms

Terms with the same letter, like 2x and 5x.

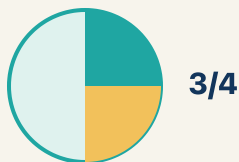

$$2 + 3 = 5$$

$$2 + 3 = 5$$

8m + 3m = 11m — add the coefficients (8 + 3 = 11) and keep the variable (m)

Combine

To add or subtract terms with the same letter into one.



2a + 5a + 3 simplifies to 7a + 3 — the two a-terms combine, the constant stays

Simplify

To write an expression in its shortest form.


$$4x$$

coefficient = 4

In 9x, the coefficient is 9 — it tells you 'nine groups of x'; if x = 2, then 9x = 18

Coefficient

The number in front of a letter, like the 3 in 3x.


$$3x + 7 + y$$

3 separate terms

In $3x + 5 - 2x$, the terms are $3x$, 5 , and $2x$.

Term

A single number, variable, or product of them, separated by + or – signs.


$$3(x + 2)$$

$$3x + 6$$

$$3(x + 4) = 3 \cdot x + 3 \cdot 4 = 3x + 12.$$

Distributive property

Multiplying a sum by a number means multiplying each part by that number.

Key Ideas & Notes



WHAT WE'RE LEARNING TODAY

I can simplify algebraic expressions by combining like terms.

1 Notice

The music studio is placing supply orders for three rehearsal rooms.

2 Key idea

I can simplify algebraic expressions by combining like terms.

3 Words I'll use

Like Terms

Combine

Simplify

Coefficient

Term

Distributive property



Think About It

- Why can you add the microphones together but not mix microphones with stands?
- What makes microphones and stands 'different types' of items?
- How is this like combining like terms in algebra?

See the notes in action: watch one worked all the way through, then try the next with the same steps.



I do — watch

Follow each step as your teacher solves it.

Problem: Simplify $7x + 3x$.

A. $10x$

B. $10x^2$

C. $21x$

D. $73x$

Step 1

$7x + 3x = 10x$.

Step 2

Add the coefficients: $7 + 3 = 10$.



Answer: A. $10x$

 We do — together

Solve this one with your class using the same steps.

Problem: Simplify $9n + 4 + 2n$.

A. $11n + 4$

B. $15n$

C. $11n^2 + 4$

D. $92n + 4$

Step 1

Step 2

Answer:

 You do — your turn

Now try one on your own. Show every step.

Problem: Which pair are like terms?

A. $6x$ and $2x$

B. $6x$ and $6y$

C. $6x$ and 6

D. $6x$ and $2x^2$

Show your work:

Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

Three rooms need microphones and stands. Why can you add all the microphones together ($4 + 2 + 6$) but not mix microphones with stands?

Level 1 support

Start here: You can only combine items of the same type, just like like terms.

 **Need a hint?**

Sentence starters (optional)

I can add the microphones because they are the ____ type.

Puedo sumar los micrófonos porque son del ____ tipo.

I cannot mix microphones and stands because they ____.

No puedo mezclar micrófonos y soportes porque ____.

Word bank: Like Terms Combine Simplify Coefficient

Listen for: Students explain microphones combine with microphones ($4 + 2 + 6 = 12$) but not with stands, connecting this to combining like terms in algebra.

Level 2

How is mixing microphones and stands like trying to add $3x$ and 5 ? Generalize the rule for which terms can combine.

- Mixing microphones and stands is like adding $3x$ and 5 because ____.
- In general, terms combine only when they have the ____.

EXPLORE

To simplify $6x + 4 + 2x + 7$, which are the like terms, and how do you combine them?

Level 1 support

Start here: Combine the x-terms by adding coefficients; combine the constants separately.



Need a hint?

Sentence starters (optional)

The like terms are ____ and ____.

Los términos semejantes son ____ y ____.

I combine them by ____ to get ____.

Los combino al ____ para obtener ____.

Word bank:

Like Terms

Combine

Simplify

Coefficient

Listen for: A strong answer combines $6x + 2x = 8x$ and $4 + 7 = 11$ to get $8x + 11$, adding coefficients of like terms and keeping the variable.

Level 2

A classmate simplifies $5x + 3 + 2y + 4x$ as $11xy + 3$. Critique the error and give the correct simplified form.

- The error is ____ because $5x$ and $2y$ are not ____.
- The correct simplified form is ____ because I combine only ____.

CONNECT

The festival order is $(8m + 5) + (3m + 7) + (4m + 2)$, where m is microphones per stage. What does the simplified expression tell you?

Level 1 support

Start here: Combine the m -terms and the constants to read the total at a glance.

 **Need a hint?**

Sentence starters (optional)

The total simplifies to ____ m + ____.

El total se simplifica a ____ m + ____.

The ____ m tells us ____ and the constant tells us ____.

El ____ m nos dice ____ y la constante nos dice ____.

Word bank: Like Terms Combine Simplify Coefficient

Listen for: Students simplify to $15m + 14$, explaining $15m$ is the total cables tied to microphones and 14 is the extra fixed cables across stages.

Level 2

Why is $15m + 14$ easier to use for ordering than the original three grouped expressions? Justify how simplifying makes the real problem clearer.

- The simplified form is easier because ____.
- If each stage had 6 microphones, the total cables would be ____, which is faster to find because ____.

Write About the Math

The Writing Revolution

I can explain my steps using the words like terms, combine, simplify, and coefficient.

1. Kernel Sentence subject + verb

Model: Simplify is to write an expression in its shortest form.

Simplificar es escribir una expresión en su forma más corta.

Write a kernel sentence about simplify. Use a subject and a verb.

Escribe una oración base sobre simplificar. Usa un sujeto y un verbo.

2. Sentence Expansion because · but · so

Kernel: Simplify matters in math

Simplificar importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Simplify matters in math because ____.

Simplificar importa en matemáticas porque ____.

but
pero

Simplify matters in math, but ____.

Simplificar importa en matemáticas, pero ____.

so
entonces

Simplify matters in math, so ____.

Simplificar importa en matemáticas, entonces ____.

3. Sentence Types 4 ways to write a math idea

Statement
Afirmación

Tell one true fact about simplify.
Di un hecho verdadero sobre simplify.

Simplify ____.

Question
Pregunta

Ask a question about simplify.
Haz una pregunta sobre simplify.

How does ____ ?

¿Cómo ____ ?

Exclamation
Exclamación

Show excitement about simplify.
Muestra entusiasmo sobre simplify.

Wow, ____ !

¡Guau, ____ !

Command
Mandato

Tell a partner what to do with simplify.
Dile a un compañero qué hacer con simplify.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning use a sentence starter

I can add the microphones because they are the ____ type.

Puedo sumar los micrófonos porque son del ____ tipo.

I cannot mix microphones and stands because they ____.

No puedo mezclar micrófonos y soportes porque ____.

The like terms are ____ and ____.

Los términos semejantes son ____ y ____.

Try It

Solve on your own. Check the answer key when you are done.

1. Simplify: $3(x + 2) + 4x$.

A. $7x + 6$

B. $3x + 6$

C. $7x + 2$

D. $12x$

Show your work:

2. Simplify: $8a + 4 - 3a - 1$.

A. $5a + 3$

B. $5a + 5$

C. $11a + 3$

D. $5a - 3$

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

Three students simplified $4x + 3 + 2x + 5 + x$ differently. Student A got $7x + 8$. Student B got $7x^2 + 8$. Student C got $6x + 9$. Who is correct? Explain the mistakes the other two students made.

Sentence starter: Student ____ is correct because $4x + 2x + x = \underline{\hspace{1cm}}x$ and $3 + 5 = \underline{\hspace{1cm}}$. Student ____'s error was _____. Student ____'s error was _____.

Show your work:

Reflect — Exit Ticket

Simplify $6x + 3 + 2x + 5$.

- A. $8x + 8$
- B. $8x^2 + 8$
- C. $62x + 35$
- D. $16x$

Your answer:

Answer Key & Teacher Guide

1. **We Try (notes):** A. $11n + 4 - 9n + 2n = 11n$. *The constant 4 has no like term, so it stays.*
Result: $11n + 4$.
2. **You Try (notes):** A. $6x$ and $2x - 6x$ and $2x$ are like terms because they both have the same variable (x) raised to the same power (1).
3. **Try It 1:** A. $7x + 6 - 3x + 6 + 4x = 7x + 6$.
4. **Try It 2:** A. $5a + 3 - 8a - 3a = 5a$; $4 - 1 = 3$; result $5a + 3$.
5. **Exit Ticket:** A. $8x + 8 - 6x + 2x = 8x$ and $3 + 5 = 8$, so the simplified expression is $8x + 8$.

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Simplify is to write an expression in its shortest form.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).